ITSC 206: Advanced Network Infrastructure & Security

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Project Focus: Multi-Firewall Routing, Active Directory Integration, and Network Segmentation

Project Overview

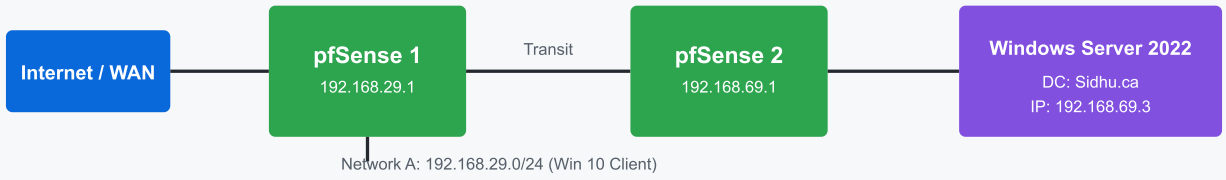
This technical report documents the configuration and integration of a multi-subnet enterprise network environment. The primary objective was to deploy and configure two pfSense firewalls (pfSense 1 and pfSense 2) to manage secure traffic flow between distinct network segments, deploy a Windows Server 2022 Domain Controller, and verify client connectivity across routed boundaries.

The labs progressively built a secure infrastructure, moving from foundational DHCP and basic static IP assignments to complex Active Directory domain joining across subnets, static routing, and granular firewall rule enforcement using port and IP aliases.

> **Key Achievement:** Successfully joined a Windows 10 client (Network A) to a Windows Server 2022 Active Directory domain (Network C) by traversing two pfSense firewalls using static routes and custom DNS forwarding.

Network Architecture

The network is segmented into distinct zones, relying on pfSense to handle NAT, DHCP, and inter-VLAN routing. The diagram below illustrates the logical topology established in Lab 5 and Lab 6.



Firewall Configuration & Alias Management

To maintain a clean and scalable security posture, custom aliases were created on both pfSense instances. This allowed rules to reference logical names rather than raw IP addresses or port numbers, streamlining the configuration of Active Directory communication and general network traffic.

Key Port Aliases (pfSense 1 & pfSense 2)

Alias Name	Type	Values	Description
AD	Port(s)	135, 88, 389, 53, 445	Active Directory required ports
allow_ssh	Port(s)	2222, 7234, 8456, 9123, 4567	Custom SSH / knock ports
internet	Port(s)	80, 443, 53	Standard web and DNS traffic

pfSense 1 LAN Rules (Network A)

The LAN rules were strictly configured to permit essential services while implicitly dropping unauthorized traffic. Specific rules were created to allow dynamic RPC ports and Active Directory traffic to reach the Windows Server located in Network C.

Action	Protocol	Source	Port	Destination	Port	Description
Pass	*	*	*	LAN Address	443, 80, 22	Anti-Lockout Rule
Pass	IPv4 TCP/UDP	Network_A	*	192.168.69.3	49152-65535	Allow dynamic RPC ports
Pass	IPv4 TCP/UDP	Network_A	*	192.168.69.3	AD	Allow AD to Network C
Pass	IPv4 ICMP	Network_A	*	*	*	Allow ping to any
Pass	IPv4 TCP/UDP	Network_A	*	*	internet	Allow internet through LAN
Block	IPv4 *	*	*	*	*	Block any other protocols

Static Routing & Inter-Subnet Connectivity

Because the network contains multiple routers (pfSense 1 and pfSense 2), static routing was necessary to inform each firewall how to reach the subnets behind the other. Gateways were defined on both appliances, and static routes were mapped to ensure seamless end-to-end communication between the Windows 10 client (Network A) and the Windows Server (Network C).

pfSense Static Routes

Firewall	Network	Gateway	Interface	Description
pfSense 1	192.168.69.0/24	WAN_DHCP - 10.29.0.20	WAN	Route to Network C (Server)
pfSense 2	192.168.29.0/24	GW_PF1 - 10.29.0.21	OPT1	Route to Network A (Clients)

Active Directory & DNS Configuration

The Windows Server 2022 instance (`WIN-TTRTB8DT4VT`) was configured as the primary Domain Controller for the **Sidhu.ca** domain. It was assigned a static IP address of `192.168.69.3`, placed intentionally outside the pfSense 2 DHCP pool (192.168.69.10 - 100) to prevent IP conflicts.

To allow domain clients to access the internet while still resolving internal Active Directory queries, DNS Forwarding was configured on the Windows Server. The DNS Server was set to forward unresolved queries (such as external web traffic) to Google's public DNS servers (`8.8.8.8`), ensuring robust network resolution across the topology.

> **pfSense 2 LAN Rule Addition:** A specific firewall rule was added to pfSense 2 to permit IPv4 UDP traffic from the server (`192.168.69.3`) on port 53 (DNS), explicitly allowing this DNS forwarding traffic to pass securely.

Client Connectivity and Verification

The final phase of the lab involved configuring a Windows 10 client machine to validate the DHCP scope, routing tables, firewall rules, and Active Directory authentication. The client successfully received its IP configuration dynamically from pfSense, with the DNS server explicitly pointing to the Domain Controller.

The output below confirms the client successfully obtained an IP in Network A (`192.168.29.12`), routed through the default gateway (`192.168.29.1`), and authenticated to the `Sidhu.ca` domain hosted on `192.168.69.3`.

```
      Microsoft Windows [Version 10.0.19044.1288] (c) Microsoft Corporation. All
rights reserved. C:\Users\ksidhu>systeminfo | findstr /B "Domain" Domain: Sidhu.ca
C:\Users\ksidhu>ipconfig /all Windows IP Configuration Host Name . . . . . .
. : DESKTOP-MP2840U Primary Dns Suffix . . . . . : Sidhu.ca Ethernet adapter
Ethernet0: Connection-specific DNS Suffix . : home.arpa IPv4 Address. . . . . .
. . : 192.168.29.12(Preferred) Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.29.1 DHCP Server . . . . . :
192.168.29.1 DNS Servers . . . . . : 192.168.69.3
```

Conclusion & Key Learnings

This deployment successfully demonstrated the integration of open-source firewall routing with enterprise Microsoft environments. The critical takeaways include the importance of precise static routing when dealing with multiple NAT gateways, the power of pfSense aliases in maintaining readable and secure firewall rulesets, and the absolute necessity of correct DNS flow (Client → DC → External Forwarder) when integrating Active Directory across segmented networks.